

Cross-sectional factor backtester: methodology note

US large caps, 2010-2026. Generated 2026-09-11 from the saved runs; the repository README carries the same numbers.

Question

Do value, momentum, quality and low volatility earn a premium on a universe an outsider can verify, once the fundamentals are lagged to their filing date, costs and a one-day execution lag are charged, and the number of specifications tried is counted honestly?

Data, and what is honest about it

- Universe: S&P 500 membership reconstructed month by month from Wikipedia's changes table, cross-checked against an independent daily list (98.4% of universe-months agree, every disagreement listed) and 20 changes checked against S&P press releases. 496-504 names at every month-end, 818 unique names over the window.
- Fundamentals: SEC Financial Statement Data Sets, 2009q1-2026q2, 16 concepts through an ordered XBRL tag map. Every value is keyed on its filing date; the first-filed value wins over amendments; the value used at month-end t was filed on or before t - 1 day. This join has three unit tests, one of which plants a filing dated after the signal date and asserts it is excluded.
- Prices: yfinance daily adjusted closes. Delisted names are largely absent: 14.6% of universe-months have no price, 31% in 2010 falling to 0% today. yfinance's series for reused symbols are dropped by three logged rules (wrong entity, bad print, corrupt series).
- Benchmarks: Ken French's factors and the six size x B/M and size x profitability portfolios, whose big-cap legs are the like-for-like comparison for a large-cap universe.

Method

- Signals at each month-end from data available then: 12-1 momentum, book-to-price, earnings yield, gross profit / assets, cash-flow accruals (negated), 252-day volatility (negated). Winsorised at the 1st/99th percentile, z-scored within sector (SIC mapped by hand to 11 GICS-like buckets), composites averaged and re-standardised.
- Portfolios: ten equal-count deciles, equal-weighted (cap-weighted as a sensitivity), formed monthly and traded at the close of the next trading day; long-short is decile 10 minus 1. Turnover is half the sum of absolute weight changes against drifted weights, both legs. Every dollar bought or sold pays the one-way cost c, so $r_{net} = r_{gross} - 2c \times \text{turnover}$; the brief's $r_{gross} - c \times \text{turnover}$ charges half that.
- Statistics: Spearman IC by month and by horizon 1-12 with an exponential half-life; Fama-MacBeth premia with Newey-West errors (checked against statsmodels to 1e-9); attribution on Mkt-RF, HML, UMD, RMW; break-even cost; and the deflated Sharpe with N read from a specification log that every run appends to (N = 65).

Results (net of 10 bp; DSR uses N = 65)

Factor	Gross	Net	Vol	Sharpe	DSR	MaxDD	Turn.	IC	IC t	Break-even
Momentum 12-1	2.2%	0.7%	15.8%	0.05	0.01	-56%	0.62	0.007	0.6	15 bp
Value (B/P, E/P)	5.3%	4.6%	9.2%	0.50	0.35	-22%	0.28	0.011	1.4	80 bp
Quality (GP/A, accruals)	2.1%	1.4%	8.5%	0.17	0.04	-24%	0.26	0.009	1.5	33 bp
Low volatility	-4.1%	-4.7%	18.9%	-0.25	0.00	-71%	0.24	0.003	0.2	none
Composite	5.9%	5.0%	14.5%	0.34	0.15	-28%	0.40	0.018	1.9	62 bp

Validation

Momentum long-short correlates 0.79 with UMD (0.85 without sector neutralisation) and loads on it with beta 0.91, R-squared 0.66: it is UMD, and UMD earned nothing in this window. The reported value and quality factors are sector-neutral composites and correlate 0.25 and 0.07 with HML and RMW, which is construction, not a join error. The join is tested by replicating French's construction (one signal, no sector neutralisation, cap-weighted terciles) against the big-cap half of his factor: the B/P replication correlates 0.74 with the big-cap HML leg over the window and 0.89 from 2016. The profitability replication reaches 0.44, rising from 0.07 in 2010-12 to 0.74 in 2019-21 as XBRL coverage does; it is reported as a fail.

What the numbers mean

- Value worked (net Sharpe 0.50, Fama-MacBeth t 3.2, break-even 80 bp). Its 4.9% attribution alpha is the number to distrust first: sector-neutral B/P alone has 4.5% (t 2.3) with an HML loading of 0.26, so the alpha is what sector neutralisation leaves after a factor that is not sector-neutral. Cap-weighted it is 0.30; on the well-covered months from 2015-12 it is 0.48.
- Momentum is marginal after costs (0.05; break-even 15 bp at 0.62 monthly turnover) and better held 3-12 months (0.18-0.23). Quality is small (0.17; 0.42 cap-weighted). Low volatility loses as a long-short: beta -0.65 (t -10.5), RMW 0.94 (t 8.2), alpha 1.1% (t 0.4) - it is a short-beta position and shorting beta lost for sixteen years.
- Signal decay: momentum's IC halves in about 15 months; value, quality and the composite do not decay within 12 months, so holding them 6-12 months keeps the return and cuts most of the turnover (composite: 0.34 monthly, 0.64 at 12

months).

- The deflated Sharpe of the composite is 0.15: a 15% probability that its Sharpe beats the best of 65 random trials with the same dispersion. Nothing here is a 2.0 Sharpe.

What did not work

- Wikipedia's 'Date added' is a corporate-event date for long-standing members (Sempra 2017, T. Rowe Price 2019); trusted only before 2010, adjudicated by the cross-check after.
- Stooq serves a JavaScript challenge, not data; yfinance stitches reused symbols and carries bad prints. Delisted names remain the survivorship that is left, and it is quantified.
- The first momentum run correlated 0.04 with UMD: uncleaned prices and a validation join off by one month. Both runs stay in the specification log.
- Wikipedia's CIK for ExxonMobil is a 2026 entity; some filers report share counts in the wrong units; OperatingIncomeLoss is not reported by banks. Each has a logged rule.

Code, tests, the specification log and every check file are in the repository. The results table, charts and this note are regenerated by 'backtester report'.